

Wykrywanie zagrożeń i reakcja na incydenty

Laboratorium nr 3 Grupa 5, śr 15:15

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Część 1: Zbieranie pakietów

Skonfigurowano dwie maszyny wirtualne: Kali Linux oraz Ubuntu Server. Na obu systemach sprawdzono dostępne interfejsy sieciowe przy użyciu poleceń `ifconfig` oraz `ip link`.

Zidentyfikowano interfejsy sieciowe oraz przypisane im adresy IP. Następnie sprawdzono komunikację między maszynami za pomocą polecenia `ping` - połączenie działa poprawnie w obie strony. Na serwerze Ubuntu uruchomiono usługi SSH oraz FTP i zweryfikowano ich działanie. Z maszyny Kali nawiązano połączenia do serwera przy użyciu SSH oraz FTP, co potwierdziło poprawną konfigurację usług oraz komunikacji sieciowej.

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:35:91:bd brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.3/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 547sec preferred_lft 547sec
    inet6 fe80::a00:27ff:fe35:91bd/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

Rys. 1 Sprawdzenie interfejsów sieciowych na Ubuntu Server (`ip a`).

```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 08:00:27:77:d3:56 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.4 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe8e:6425 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8e:64:25 txqueuelen 1000 (Ethernet)
    RX packets 12 bytes 3587 (3.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 31 bytes 3676 (3.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$
```

Rys. 2 Sprawdzenie interfejsów sieciowych na Kali Linux (`ifconfig`).

```
(kali@kali)-[~]
$ ping 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=0.486 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=0.291 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=1.00 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=0.848 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=0.608 ms
^C
— 10.0.2.3 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4058ms
rtt min/avg/max/mdev = 0.291/0.646/1.001/0.252 ms

(kali@kali)-[~]
$
```

Rys. 3 Ping z Kali Linux do Ubuntu Server - poprawne połączenie.

```
ubuntu@ubuntu:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=0.906 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=0.486 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=0.374 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=0.431 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=0.903 ms
^C
--- 10.0.2.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.374/0.620/0.906/0.234 ms
```

Rys. 4 Ping z Ubuntu Server do Kali Linux - poprawne połączenie.

```
(kali@kali)-[~]
$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:77:d3:56 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:8e:64:25 brd ff:ff:ff:ff:ff:ff
```

Rys. 5 Wyświetlenie interfejsów sieciowych na Kali Linux (ip link).

```
ubuntu@ubuntu:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:35:91:bd brd ff:ff:ff:ff:ff:ff
ubuntu@ubuntu:~$
```

Rys. 6 Wyświetlenie interfejsów sieciowych na Ubuntu Server.

```
ubuntu@ubuntu:~$ sudo systemctl status ssh
[sudo] password for ubuntu:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-04-10 06:07:18 UTC; 3min 5s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1123 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 1124 (sshd)
    Tasks: 1 (limit: 4601)
   Memory: 4.0M (peak: 4.9M)
      CPU: 159ms
   CGroup: /system.slice/ssh.service
           └─1124 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Rys. 7 Sprawdzenie działania usługi SSH - usługa aktywna.

```

ubuntu@ubuntu:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-04-10 06:03:25 UTC; 8min ago
     Process: 596 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
    Main PID: 613 (vsftpd)
      Tasks: 1 (limit: 4601)
     Memory: 928.0K (peak: 1.4M)
        CPU: 21ms
    CGroup: /system.slice/vsftpd.service
           └─613 /usr/sbin/vsftpd /etc/vsftpd.conf

```

Rys. 8 Sprawdzenie działania usługi FTP - usługa aktywna.

```

(kali@kali)-[~]
$ ssh ubuntu@10.0.2.3
ubuntu@10.0.2.3's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 10 06:13:14 AM UTC 2026

System load:  0.02               Processes:    111
Usage of /:   13.3% of 24.44GB   Users logged in: 1
Memory usage: 7%                IPv4 address for enp0s3: 10.0.2.3
Swap usage:  0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

73 updates can be applied immediately.
56 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Apr 10 06:07:25 2026 from 10.0.2.1
ubuntu@ubuntu:~$

```

Rys. 9 Połączenie SSH z Kali do Ubuntu zakończone sukcesem.

```

(kali@kali)-[~]
$ ftp 10.0.2.3
Connected to 10.0.2.3.
220 (vsFTPd 3.0.5)
Name (10.0.2.3:kali): ubuntu
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>

```

Rys. 10 Połączenie FTP z Kali do Ubuntu zakończone sukcesem.

Polecenia tcpdump i tcpflow służą do analizy ruchu sieciowego.

tcpdump przechwytuje pakiety i pokazuje ruch w sieci w czasie rzeczywistym.

tcpflow pozwala zobaczyć pełne dane przesyłane między komputerami.

Najważniejsze opcje to wybór interfejsu (-i), zapis do pliku (-w) oraz filtrowanie ruchu np. po adresie IP lub porcie.

Uruchomiono narzędzie tcpdump na wybranym interfejsie sieciowym w celu przechwytywania pakietów ICMP. Ustawiono parametry umożliwiające zapis pakietów do pliku oraz ich analizę w formacie liczbowym i binarnym. Podczas działania programu przechwycono pakiety generowane przez polecenie ping. Zarejestrowany ruch zapisano do pliku pcap-ping, który może być wykorzystany do dalszej analizy.

```
(kali@kali)-[~]
$ ping -c 1024 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=0.338 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=0.392 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=0.334 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=0.852 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=0.562 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=0.567 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=0.777 ms
64 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=0.358 ms
64 bytes from 10.0.2.3: icmp_seq=9 ttl=64 time=0.441 ms
64 bytes from 10.0.2.3: icmp_seq=10 ttl=64 time=0.620 ms
64 bytes from 10.0.2.3: icmp_seq=11 ttl=64 time=0.975 ms
64 bytes from 10.0.2.3: icmp_seq=12 ttl=64 time=0.725 ms
64 bytes from 10.0.2.3: icmp_seq=13 ttl=64 time=0.678 ms
64 bytes from 10.0.2.3: icmp_seq=14 ttl=64 time=0.920 ms
64 bytes from 10.0.2.3: icmp_seq=15 ttl=64 time=0.480 ms
64 bytes from 10.0.2.3: icmp_seq=16 ttl=64 time=0.347 ms
64 bytes from 10.0.2.3: icmp_seq=17 ttl=64 time=0.280 ms
64 bytes from 10.0.2.3: icmp_seq=18 ttl=64 time=0.383 ms
64 bytes from 10.0.2.3: icmp_seq=19 ttl=64 time=0.971 ms
64 bytes from 10.0.2.3: icmp_seq=20 ttl=64 time=0.697 ms
64 bytes from 10.0.2.3: icmp_seq=21 ttl=64 time=0.422 ms
^C
--- 10.0.2.3 ping statistics ---
21 packets transmitted, 21 received, 0% packet loss, time 20368ms
rtt min/avg/max/mdev = 0.280/0.577/0.975/0.220 ms

(kali@kali)-[~]
$ ping -c 100 -s 1000 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 1000(1028) bytes of data.
1008 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=0.284 ms
1008 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=1.13 ms
1008 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=1.19 ms
1008 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=1.15 ms
1008 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=0.308 ms
1008 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=0.547 ms
1008 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=0.285 ms
1008 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=1.29 ms
1008 bytes from 10.0.2.3: icmp_seq=9 ttl=64 time=0.538 ms
1008 bytes from 10.0.2.3: icmp_seq=10 ttl=64 time=0.537 ms
1008 bytes from 10.0.2.3: icmp_seq=11 ttl=64 time=0.887 ms
^C
--- 10.0.2.3 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10143ms
rtt min/avg/max/mdev = 0.284/0.740/1.287/0.376 ms
```

Rys. 11 Wykonanie polecenia ping z maszyny Kali Linux do serwera Ubuntu - widoczna poprawna odpowiedź.

```
(kali@kali)-[~]
$ ping -c 100 -t 100 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=0.425 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=0.273 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=0.311 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=0.342 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=0.611 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=0.548 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=0.524 ms
64 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=1.11 ms
64 bytes from 10.0.2.3: icmp_seq=9 ttl=64 time=0.607 ms
64 bytes from 10.0.2.3: icmp_seq=10 ttl=64 time=0.351 ms
64 bytes from 10.0.2.3: icmp_seq=11 ttl=64 time=0.411 ms
64 bytes from 10.0.2.3: icmp_seq=12 ttl=64 time=0.391 ms
64 bytes from 10.0.2.3: icmp_seq=13 ttl=64 time=0.299 ms
64 bytes from 10.0.2.3: icmp_seq=14 ttl=64 time=0.612 ms
64 bytes from 10.0.2.3: icmp_seq=15 ttl=64 time=0.361 ms
64 bytes from 10.0.2.3: icmp_seq=16 ttl=64 time=0.967 ms
64 bytes from 10.0.2.3: icmp_seq=17 ttl=64 time=0.663 ms
64 bytes from 10.0.2.3: icmp_seq=18 ttl=64 time=0.530 ms
64 bytes from 10.0.2.3: icmp_seq=19 ttl=64 time=1.41 ms
^C
— 10.0.2.3 ping statistics —
19 packets transmitted, 19 received, 0% packet loss, time 18318ms
rtt min/avg/max/mdev = 0.273/0.565/1.407/0.291 ms

(kali@kali)-[~]
$
```

Rys. 12 Ciąg dalszy wyników polecenia ping - brak utraty pakietów oraz stabilne połączenie.

```
(kali@kali)-[~]
$ sudo tcpdump -i eth1 -nn -tttt -X icmp -w pcap-ping
[sudo] password for kali:
tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes

^C102 packets captured
102 packets received by filter
0 packets dropped by kernel
```

Rys. 13 Uruchomienie narzędzia tcpdump na interfejsie eth1 w celu przechwytywania pakietów ICMP oraz zapisania ich do pliku pcap.

Na maszynie Kali Linux utworzono pliki zawierające listy użytkowników oraz haseł przy użyciu poleceń nano usr.txt oraz nano pass.txt. Następnie sprawdzono ich obecność w katalogu domowym poleceniem ls. Kolejno wykonano skanowanie portu 22 serwera Ubuntu przy użyciu polecenia nmap -p 22 --script ssh-brute --script-args userdb=usr.txt,passdb=pass.txt 10.0.2.3 --host-timeout 15s. Podczas skanowania narzędzie próbowało różnych kombinacji loginów i haseł, co pozwoliło na odnalezienie poprawnych danych dostępowych. W tym samym czasie na serwerze Ubuntu uruchomiono przechwytywanie ruchu sieciowego przy użyciu polecenia sudo tcpdump -i enp0s3 -nn -tttt -X -c 25 'host 10.0.2.4 and tcp[tcpflags] & tcp-ack != 0' -w pcap-brutessh. Przechwycono pakiety TCP z ustawioną flagą ACK oraz zapisano je do pliku.

Pakiety z flagą ACK służą do potwierdzania odbioru danych w komunikacji TCP.

```
(kali@kali)-[~]
$ sudo nano usr.txt

(kali@kali)-[~]
$ sudo nano pass.txt

(kali@kali)-[~]
$ ls
backdoor.war  Documents  Music      pcap-ping  Public  Templates  Videos
Desktop      Downloads  pass.txt   Pictures   rev.php  usr.txt

(kali@kali)-[~]
$
```

Rys. 14 Wyświetlenie zawartości katalogu domowego - widoczne pliki `usr.txt` oraz `pass.txt`.

```
(kali@kali)-[~]
$ nmap -p 22 --script ssh-brute --script-args userdb=usr.txt,passdb=pass.txt 10.0.2.3 --host-timeout 15s
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 02:39 -0400
NSE: [ssh-brute] Trying username/password pair: ubuntu:ubuntu
NSE: [ssh-brute] Trying username/password pair: admin:admin
NSE: [ssh-brute] Trying username/password pair: user:user
NSE: [ssh-brute] Trying username/password pair: test:test
NSE: [ssh-brute] Trying username/password pair: kali:kali
NSE: [ssh-brute] Trying username/password pair: admin:1234
NSE: [ssh-brute] Trying username/password pair: user:1234
NSE: [ssh-brute] Trying username/password pair: test:1234
NSE: [ssh-brute] Trying username/password pair: kali:1234
NSE: [ssh-brute] Trying username/password pair: user:admin
NSE: [ssh-brute] Trying username/password pair: test:admin
NSE: [ssh-brute] Trying username/password pair: kali:admin
NSE: [ssh-brute] Trying username/password pair: admin:password
NSE: [ssh-brute] Trying username/password pair: user:password
NSE: [ssh-brute] Trying username/password pair: test:password
NSE: [ssh-brute] Trying username/password pair: kali:password
NSE: [ssh-brute] Trying username/password pair: admin:ubuntu
NSE: [ssh-brute] Trying username/password pair: user:ubuntu
NSE: [ssh-brute] Trying username/password pair: test:ubuntu
NSE: [ssh-brute] Trying username/password pair: kali:ubuntu
NSE: [ssh-brute] Trying username/password pair: admin:kali
NSE: [ssh-brute] Trying username/password pair: user:kali
NSE: [ssh-brute] Trying username/password pair: test:kali
Nmap scan report for 10.0.2.3
Host is up (0.00044s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-brute:
|   Accounts:
|   | ubuntu:ubuntu - Valid credentials
|_  Statistics: Performed 23 guesses in 8 seconds, average tps: 2.9
MAC Address: 08:00:27:35:91:BD (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.90 seconds

(kali@kali)-[~]
$
```

Rys. 15 Wykonanie skanowania portu 22 przy użyciu narzędzia `nmap` z wykorzystaniem skryptu `ssh-brute`.

```
ubuntu@ubuntu:~$ sudo tcpdump -i enp0s3 -nn -tttt -X -c 25 'host 10.0.2.4 and tcp[tcpflags] & tcp-ack != 0' -w pcap-b
[sudo] password for ubuntu:
tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
25 packets captured
186 packets received by filter
0 packets dropped by kernel
ubuntu@ubuntu:~$
```

Rys. 16 Uruchomienie `tcpdump` na serwerze Linux w celu przechwycenia pakietów TCP z flagą ACK oraz zapis do pliku.

Na maszynie Kali Linux przeprowadzono skanowanie serwera Ubuntu trzema metodami: SYN, NULL oraz XMAS przy użyciu poleceń `nmap -sS 10.0.2.3`, `nmap -sN 10.0.2.3` oraz `nmap -sX 10.0.2.3`. Każdy ze skanów pozwala na identyfikację stanu portów w inny sposób. W wyniku skanowania uzyskano informacje o otwartych oraz filtrowanych portach na serwerze. W trakcie wykonywania skanów na serwerze Ubuntu uruchomiono przechwytywanie ruchu sieciowego przy użyciu polecenia `sudo tcpdump -i enp0s3 -nn -tttt -X -w pcap-syn`. Przechwycone pakiety zostały zapisane do pliku w celu dalszej analizy.

```
(kali@kali)-[~]
$ nmap -sS 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 02:46 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00019s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:35:91:BD (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.81 seconds

(kali@kali)-[~]
$ nmap -sN 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 02:46 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00033s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
80/tcp    open|filtered http
MAC Address: 08:00:27:35:91:BD (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.94 seconds

(kali@kali)-[~]
$ nmap -sX 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 02:47 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0013s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
80/tcp    open|filtered http
MAC Address: 08:00:27:35:91:BD (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.96 seconds

(kali@kali)-[~]
$
```

Rys. 17 Wykonanie skanowania typu SYN, NULL oraz XMAS przy użyciu narzędzia nmap.

```
ubuntu@ubuntu:~$ sudo tcpdump -i enp0s3 -nn -tttt -X -w pcap-syn
tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C6019 packets captured
6020 packets received by filter
0 packets dropped by kernel
ubuntu@ubuntu:~$
```

Rys. 18 Przechwytywanie ruchu sieciowego przy użyciu tcpdump podczas wykonywania skanu SYN oraz zapis do pliku.

W celu weryfikacji poprawności zapisanych plików przechwyconego ruchu sieciowego użyto polecenia `file pcap-*`. System poprawnie rozpoznał pliki jako pliki typu `pcap` zawierające zapis pakietów sieciowych.

```
ubuntu@ubuntu:~$ file pcap-*
pcap-brutessh: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 262144)
pcap-syn:      pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 262144)
ubuntu@ubuntu:~$ ^C
ubuntu@ubuntu:~$
```

Rys. 19 Sprawdzenie typu plików przechwyconego ruchu sieciowego przy użyciu polecenia `file`.

Na serwerze Ubuntu uruchomiono narzędzie `p0f` przy użyciu polecenia `sudo p0f -i enp0s3`. Program służy do pasywnej analizy ruchu sieciowego oraz identyfikacji systemów operacyjnych hostów na podstawie przechwyconych pakietów. Podczas działania narzędzia wykryto host Kali Linux oraz określono jego przybliżony system operacyjny i parametry połączenia. Wyniki działania programu zapisano do pliku przy użyciu polecenia `sudo p0f -i enp0s3 > p0f.txt`. Równocześnie na maszynie Kali przechwytywano ruch sieciowy poleceniem `sudo tcpdump -i eth1 -nn -tttt -X host 10.0.2.4 -w pcap-p0f`, filtrując pakiety po adresie IP hosta generującego ruch. W pliku `p0f.txt` znajdują się informacje o adresach IP, wykrytym systemie operacyjnym, portach oraz parametrach połączenia. Plik `pcap-p0f` zawiera natomiast surowe pakiety sieciowe. Porównując oba pliki można zauważyć, że `p0f.txt` zawiera już gotową analizę ruchu, natomiast plik `pcap-p0f` zawiera dane do dalszej analizy. Informacje w tych plikach nie są identyczne, lecz się uzupełniają. Do dodatkowej analizy ruchu wykorzystano również narzędzia `tcpflow` oraz `tcpick`, które umożliwiają analizę połączeń TCP oraz odtwarzanie przesyłanych danych.

```
(kali@kali)-[~]
$ ping 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data:
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=0.382 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=0.555 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=0.927 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=0.727 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=0.439 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=0.507 ms
^C
--- 10.0.2.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5041ms
rtt min/avg/max/mdev = 0.382/0.589/0.927/0.185 ms

(kali@kali)-[~]
$ ssh ubuntu@10.0.2.3
ubuntu@10.0.2.3's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 10 06:57:36 AM UTC 2026

System load:  0.16               Processes:    124
Usage of /:   13.1% of 24.44GB   Users logged: 1
Memory usage: 10%               IPv4 address for enp0s3: 10.0.2.3
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
2 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Apr 10 06:13:15 2026 from 10.0.2.4
ubuntu@ubuntu:~$ exit
logout
Connection to 10.0.2.3 closed.

(kali@kali)-[~]
$
```

Rys. 20 Widok terminala przedstawiający połączenia sieciowego przy użyciu polecenia `ping` oraz nawiązanie połączenia SSH między maszynami.


```

ubuntu@ubuntu:~$ sudo p0f -i enp0s3
--- p0f 3.09b by Michal Zalewski <lcamtuf@coredump.cx> ---

[+] Closed 1 file descriptor.
[+] Loaded 322 signatures from '/etc/p0f/p0f.fp'.
[+] Intercepting traffic on interface 'enp0s3'.
[+] Default packet filtering configured [+VLAN].
[+] Entered main event loop.

.-[ 10.0.2.4/38362 -> 10.0.2.3/22 (syn) ]-
|
| client    = 10.0.2.4/38362
| os        = Linux 2.2.x-3.x
| dist      = 0
| params    = generic tos:0x2e
| raw_sig   = 4:64+0:0:1460:mss*44,9:mss,sok,ts,nop,ws:df,id+:0
|
|-----
.-[ 10.0.2.4/38362 -> 10.0.2.3/22 (mtu) ]-
|
| client    = 10.0.2.4/38362
| link      = Ethernet or modem
| raw_mtu   = 1500
|
|-----
.-[ 10.0.2.4/38362 -> 10.0.2.3/22 (syn+ack) ]-
|
| server    = 10.0.2.3/22
| os        = ???
| dist      = 0
| params    = none
| raw_sig   = 4:64+0:0:1460:mss*45,7:mss,sok,ts,nop,ws:df:0
|
|-----
.-[ 10.0.2.4/38362 -> 10.0.2.3/22 (mtu) ]-
|
| server    = 10.0.2.3/22
| link      = Ethernet or modem
| raw_mtu   = 1500
|
|-----

```

Rys. 21 Widok terminala systemu Ubuntu wykorzystywanego do przeprowadzenia pasywnej analizy ruchu sieciowego przy użyciu narzędzia p0f.

Część 2: t-shark oraz Wireshark

Podstawy pracy z t-sharkiem i wiresharkiem

T-shark i wireshark zostały uruchomione na maszynie wirtualnej kali, nasłuchiwały na interfejsie eth0. Poniżej znajdują się zrzuty każdej z program. Wireshark ma interfejs graficzny, w którym można zmieniać w sposób intuicyjny jakie szczegóły pakietów mają być widoczne i ile miejsca na ekranie mają przeznaczone. Poszczególne pakiety można otworzyć w każdym momencie nasłuchiwania albo po zakończeniu/zapisaniu i ponownym otwarciu pliku zawierającym ruch sieciowy. Natomiast t-shark pozwala szybko i automatycznie analizować ruch sieciowy w terminalu bez użycia interfejsu graficznego.

```
l--$ tshark -i 1
Capturing on 'eth0'
17:44:11.26985482 fe80::20c:29ff:feef:4d77 → ff02::2 Router Solicitation
17:44:12.63401932 192.168.122.128 → 192.168.122.2 47808 53 Standard query 0xe77c A google.com
17:44:12.63406632 192.168.122.128 → 192.168.122.2 47808 53 Standard query 0x6279 AAAA google.com
17:44:12.64342612 192.168.122.2 → 192.168.122.128 53 47808 Standard query response 0xe77c A google.com A 142.250.120.101 A 142.250.120.100
A 142.250.120.102 A 142.250.120.113 A 142.250.120.139 A 142.250.120.138
17:44:12.68480072 192.168.122.2 → 192.168.122.128 53 47808 Standard query response 0x6279 AAAA google.com AAAA 2a00:1450:4025:802::8a AAAA
2a00:1450:4025:802::8b AAAA 2a00:1450:4025:802::66 AAAA 2a00:1450:4025:802::64
17:44:12.68506862 192.168.122.128 → 142.250.120.101 Echo (ping) request id=0x642a, seq=1/256, ttl=64
17:44:12.76331192 142.250.120.101 → 192.168.122.128 Echo (ping) reply id=0x642a, seq=1/256, ttl=128 (request in 6)
17:44:12.76349172 192.168.122.128 → 192.168.122.2 55462 53 Standard query 0x1a4a PTR 101.120.250.142.in-addr.arpa
17:44:12.84060342 192.168.122.2 → 192.168.122.128 53 55462 Standard query response 0x1a4a PTR 101.120.250.142.in-addr.arpa PTR zo-in-f101.1
e100.net
17:44:13.68713722 192.168.122.128 → 142.250.120.101 Echo (ping) request id=0x642a, seq=2/512, ttl=64
17:44:13.72121302 142.250.120.101 → 192.168.122.128 Echo (ping) reply id=0x642a, seq=2/512, ttl=128 (request in 10)
17:44:14.70544172 VMware:cd:00:08 → Nearest-Bridge MA/00:50:56:c0:00:08 MA/00:50:56:c0:00:08 3601
17:45:12.55357192 192.168.122.128 → 192.168.122.2 36826 53 Standard query 0x5ac2 A content-signature-2.cdn.mozilla.net
17:45:12.55361792 192.168.122.128 → 192.168.122.2 36826 53 Standard query 0x72c4 AAAA content-signature-2.cdn.mozilla.net
17:45:12.55458162 192.168.122.2 → 192.168.122.128 53 36826 Standard query response 0x5ac2 A content-signature-2.cdn.mozilla.net A 34.160.14
4.191
17:45:12.56370342 192.168.122.2 → 192.168.122.128 53 36826 Standard query response 0x72c4 AAAA content-signature-2.cdn.mozilla.net AAAA 260
0:19010:92a9::
17:45:12.56459442 192.168.122.128 → 34.160.144.191 48112 443 48112 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=189699637 Tse
q=0 WS=1024
17:45:12.59422792 34.160.144.191 → 192.168.122.128 443 48112 443 → 48112 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
17:45:12.59427832 192.168.122.128 → 34.160.144.191 48112 443 48112 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
17:45:12.60050872 192.168.122.128 → 34.160.144.191 48112 443 Client Hello (SNI=content-signature-2.cdn.mozilla.net)
17:45:12.60072412 34.160.144.191 → 192.168.122.128 443 48112 443 → 48112 [ACK] Seq=1 Ack=221 Win=64240 Len=0
17:45:12.64774652 34.160.144.191 → 192.168.122.128 443 48112 Server Hello, Certificate
17:45:12.64781902 192.168.122.128 → 34.160.144.191 48112 443 48112 → 443 [ACK] Seq=221 Ack=2825 Win=65535 Len=0
17:45:12.64796622 34.160.144.191 → 192.168.122.128 443 48112 Server Key Exchange, Server Hello Done
17:45:12.64797702 192.168.122.128 → 34.160.144.191 48112 443 48112 → 443 [ACK] Seq=221 Ack=3812 Win=65535 Len=0
17:45:12.65133402 192.168.122.128 → 34.160.144.191 48112 443 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
17:45:12.65171772 34.160.144.191 → 192.168.122.128 443 48112 443 → 48112 [ACK] Seq=3012 Ack=314 Win=64240 Len=0
17:45:12.67944202 34.160.144.191 → 192.168.122.128 443 48112 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Applicati
on Data
17:45:12.72141732 192.168.122.128 → 34.160.144.191 48112 443 48112 → 443 [ACK] Seq=314 Ack=3392 Win=65535 Len=0
17:45:12.73060882 192.168.122.128 → 192.168.122.2 49769 53 Standard query 0xa6f6 A ads.mozilla.org
17:45:12.73065032 192.168.122.128 → 192.168.122.2 49769 53 Standard query 0x8ef4 AAAA ads.mozilla.org
17:45:12.74095822 192.168.122.2 → 192.168.122.128 53 49769 Standard query response 0xa6f6 A ads.mozilla.org CNAME mozilla.map.fastly.net A
146.75.117.91 NS ns1.fastly.net NS ns2.fastly.net NS ns3.fastly.net NS ns4.fastly.net
17:45:12.74760912 192.168.122.2 → 192.168.122.128 53 49769 Standard query response 0x8ef4 AAAA ads.mozilla.org CNAME mozilla.map.fastly.net
AAAA 2a04:2462:80::347 NS ns1.fastly.net NS ns2.fastly.net NS ns3.fastly.net NS ns4.fastly.net
17:45:12.75798012 192.168.122.128 → 146.75.117.91 49560 443 49560 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2884813035 Tse
q=0 WS=1024
17:45:12.79342902 146.75.117.91 → 192.168.122.128 443 49560 443 → 49560 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
17:45:12.79346122 192.168.122.128 → 146.75.117.91 49560 443 49560 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
```

Rys. 1. Przykład ruchu sieciowego w t-shark

Time	Source	Destination	Source port	Destination port	Niestandardowa kolumna
17:44:12.63401932	192.168.122.128	192.168.122.2	47808	53	
17:44:12.63406632	192.168.122.128	192.168.122.2	47808	53	
17:44:12.64342612	192.168.122.2	192.168.122.128	53	47808	
17:44:12.68480072	192.168.122.2	192.168.122.128	53	47808	
17:44:12.76331382	142.250.120.101	192.168.122.128			
17:44:12.76349172	192.168.122.128	192.168.122.2	55462	53	
17:44:12.84060342	192.168.122.2	192.168.122.128	53	55462	
17:44:13.68713722	192.168.122.128	142.250.120.101			
17:44:13.72121302	142.250.120.101	192.168.122.128			
17:45:12.55357192	192.168.122.128	192.168.122.2	36826	53	
17:45:12.55361792	192.168.122.128	192.168.122.2	36826	53	
17:45:12.55458162	192.168.122.2	192.168.122.128	53	36826	
17:45:12.56370342	192.168.122.2	192.168.122.128	53	36826	
17:45:12.56459442	192.168.122.128	34.160.144.191	48112	443	
17:45:12.59422792	34.160.144.191	192.168.122.128	443	48112	
17:45:12.59427832	192.168.122.128	34.160.144.191	48112	443	
17:45:12.60050872	192.168.122.128	34.160.144.191	48112	443	
17:45:12.60072412	34.160.144.191	192.168.122.128	443	48112	
17:45:12.64774652	34.160.144.191	192.168.122.128	443	48112	
17:45:12.64781902	192.168.122.128	34.160.144.191	48112	443	
17:45:12.64796622	34.160.144.191	192.168.122.128	443	48112	
17:45:12.64797702	192.168.122.128	34.160.144.191	48112	443	
17:45:12.65133402	192.168.122.128	34.160.144.191	48112	443	
17:45:12.65171772	34.160.144.191	192.168.122.128	443	48112	
17:45:12.67944202	34.160.144.191	192.168.122.128	443	48112	
17:45:12.72141732	192.168.122.128	34.160.144.191	48112	443	
17:45:12.73060882	192.168.122.128	192.168.122.2	49769	53	

Rys. 2. Przykład ruchu sieciowego w wireshark

Na Rys. 2. wyłączone są kolumny No, Protocol, kolumna Length jest wykasowana, zostały dodane kolumny Source Port, Destination Port, czas jest w formacie UTC oraz została dodana „Niestandardowa kolumna”.

Analiza plików pcap-01, pcap-02

No.	Time	Source	Destination	Protocol	Length	Register Value (UINT16)	Info
1	0.000000	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<00>
2	0.000002	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<20>
3	0.000219	10.12.3.101	10.12.3.255	NBNS	110		Registration NB WORKGROUP<00>
11	0.764150	10.12.3.101	10.12.3.255	NBNS	110		Registration NB WORKGROUP<00>
12	0.764151	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<20>
13	0.764152	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<00>
16	1.528568	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<00>
17	1.528570	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<20>
18	1.528573	10.12.3.101	10.12.3.255	NBNS	110		Registration NB WORKGROUP<00>
19	2.292993	10.12.3.101	10.12.3.255	NBNS	110		Registration NB WORKGROUP<00>
20	2.292994	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<20>
21	2.292995	10.12.3.101	10.12.3.255	NBNS	110		Registration NB MASTERSON-PC<00>
23	3.030847	10.12.3.101	255.255.255.255	DHCP	342		DHCP Inform - Transaction ID 0x8d85e249
31	3.353955	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
33	4.118260	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
34	4.882693	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
96	73.425654	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
97	73.428591	10.12.3.101	255.255.255.255	DHCP	342		DHCP Inform - Transaction ID 0xb66f8460
312	74.187421	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
313	74.951891	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
1112	125.822638	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
1114	126.574316	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
1115	127.338709	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WPAD<00>
1461	229.517096	10.12.3.101	10.12.3.255	NBNS	92		Name query NB WORKGROUP<1b>
1462	229.517677	10.12.3.101	10.12.3.1	NBNS	104		Name query response NB 10.12.3.101

Rys. 3. Odfiltrowane pakiety DHCP oraz NBNS dla pcap-01

Protocol	Source	Source Address	Host Name	Name
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<00>,MASTERSON-PC<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<20>,MASTERSON-PC<20> (Server service)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<00>,WORKGROUP<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<00>,WORKGROUP<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<20>,MASTERSON-PC<20> (Server service)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<00>,MASTERSON-PC<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<00>,MASTERSON-PC<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<20>,MASTERSON-PC<20> (Server service)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<00>,WORKGROUP<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<00>,WORKGROUP<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<20>,MASTERSON-PC<20> (Server service)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<00>,MASTERSON-PC<00> (Workstation/Redirector)
DHCP	HewlettPacka_1c:47:ae	10.12.3.101	MasterSON-PC	
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
DHCP	HewlettPacka_1c:47:ae	10.12.3.101	MasterSON-PC	
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WPAD<00>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		MASTERSON-PC<00> (Workstation/Redirector)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		DESKTOP-AL155LR<20>
NBNS	Netgear_b6:93:f1	10.12.3.1		DESKTOP-AL155LR<20> (Server service)
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
NBNS	HewlettPacka_1c:47:ae	10.12.3.101		WORKGROUP<1b>
DHCP	HewlettPacka_1c:47:ae	10.12.3.101	MasterSON-PC	
DHCP	Netgear_b6:93:f1	10.12.3.1		
DHCP	HewlettPacka_1c:47:ae	10.12.3.101	MasterSON-PC	
DHCP	Netgear_b6:93:f1	10.12.3.1		

Rys. 4. Wykryte nazwy hostów, adresy źródłowe IP oraz adresy MAC w pliku pcap-01

Wireshark · IPv4 Statistics / Source and Destination Addresses · pcap02.pcap								
Topic / Item	Cou	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
▼ Source IPv4 Addresses	1290				0,0045	100%	2,8700	3,855
217.18.244.196	852				0,0030	66,05%	2,0700	3,855
10.12.3.101	407				0,0014	31,55%	0,8000	3,856
8.8.8.8	17				0,0001	1,32%	0,0100	7,341
185.163.45.48	7				0,0000	0,54%	0,0100	82,560
95.213.251.165	6				0,0000	0,47%	0,0100	132,469
10.12.3.1	1				0,0000	0,08%	0,0100	1,603
► Destination IPv4 Addresses	1290				0,0045	100%	2,8700	3,855

Rys. 5. Statystyka poszczególnych adresów IP według ilości pakietów w pcap-02

```

Wireshark · Śledź strumień TCP (tcp.stream eq 6) · pcap03.pcap

GET /raw HTTP/1.1
User-Agent: IP retriever
Host: myexternalip.com
Cache-Control: no-cache

HTTP/1.1 200 OK
Server: nginx/1.10.3
Date: Tue, 20 Nov 2018 17:59:09 GMT
Content-Type: text/plain
Content-Length: 15
Connection: keep-alive
Expires: Tue, 20 Nov 2018 17:59:08 GMT
Cache-Control: no-cache

173.66.146.112

```

Rys. 6. TCP stream pakietu http w pliku pcap-03

Urządzenie wysyłające żądanie identyfikuje się jako IP retriever, a serwer to nginx w wersji 1.10.3. Na podstawie tych informacji nie da się określić wersji i rodzaju systemu.

Plik pcap-04 nie zawierał pakietów Kerberos, dalszejsza analiza nie możliwa.

kerberos				
Protocol	Source	Source Address	Host Name	Name

Rys. 7. Odfiltrowane pakiety Kerberos pliku pcap-04

Część 3: Pakiety internetowe. Analiza zawartości pakietów

Zastosuj filtr wyświetlania ... <Ctrl-/>					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000000	192.168.1.145	149.154.167.99	TLSv1.2	253 Application Data
2	0.042944400	149.154.167.99	192.168.1.145	TLSv1.2	167 Application Data
3	0.043907600	192.168.1.145	149.154.167.99	TLSv1.2	203 Application Data
4	0.043907600	192.168.1.145	149.154.167.99	TLSv1.2	253 Application Data
5	0.090837400	149.154.167.99	192.168.1.145	TLSv1.2	167 Application Data
6	0.091813900	149.154.167.99	192.168.1.145	TLSv1.2	167 Application Data
7	0.092133200	192.168.1.145	149.154.167.99	TCP	60 54083 → 443 [ACK] Seq=200 Ack=114 Win=63463 Len=0
8	0.139253200	192.168.1.145	149.154.167.99	TCP	60 51988 → 443 [ACK] Seq=200 Ack=114 Win=64254 Len=0
9	0.139253200	192.168.1.145	149.154.167.99	TCP	60 51993 → 443 [ACK] Seq=150 Ack=114 Win=1022 Len=0
10	0.155005100	192.168.1.145	57.144.111.134	TLSv1.2	124 Application Data
11	0.179661300	57.144.111.134	192.168.1.145	TCP	60 443 → 52020 [ACK] Seq=1 Ack=71 Win=333 Len=0
12	0.227481900	57.144.111.134	192.168.1.145	TLSv1.2	126 Application Data
13	0.277351700	192.168.1.145	57.144.111.134	TCP	60 52020 → 443 [ACK] Seq=71 Ack=73 Win=1026 Len=0
14	0.560641900	23.201.43.108	192.168.1.105	TLSv1.2	78 Application Data
15	0.560897800	23.201.43.108	192.168.1.105	TCP	60 443 → 49738 [FIN, ACK] Seq=25 Ack=1 Win=501 Len=0
16	0.560924900	192.168.1.105	23.201.43.108	TCP	54 49738 → 443 [ACK] Seq=1 Ack=26 Win=2047 Len=0
17	0.561161800	192.168.1.105	23.201.43.108	TCP	54 49738 → 443 [FIN, ACK] Seq=1 Ack=26 Win=2047 Len=0
18	0.644714500	SamsungElect_7a:05:...	Broadcast	ARP	60 Who has 192.168.1.1? Tell 192.168.1.109
19	0.862187900	192.168.1.105	23.201.43.108	TCP	54 [TCP Retransmission] 49738 → 443 [FIN, ACK] Seq=1 Ack=26 Win=2047 Len=0
20	0.889307800	23.201.43.108	192.168.1.105	TCP	60 443 → 49738 [ACK] Seq=26 Ack=2 Win=501 Len=0
21	1.120423800	192.168.1.111	192.168.1.255	UDP	260 6537 → 6537 Len=218
22	1.320958600	192.168.1.111	192.168.1.255	UDP	260 6537 → 6537 Len=218
23	1.343170300	149.154.167.99	192.168.1.145	TLSv1.2	183 Application Data
24	1.344668600	192.168.1.145	149.154.167.99	TLSv1.2	253 Application Data

Rys. 8. Przykład wygenerowanych danych na maszynie Windows.

Nie udało się odczytać ciasteczek HTTP w Wireshark, ponieważ większość ruchu odbywała się przez HTTPS, gdzie dane (w tym cookies) są zaszyfrowane i niewidoczne w przechwyconych pakietach. W Windowsie ciasteczka są przechowywane w
C:\Users\nazwa\AppData\Local\Google\Chrome\User Data\Default\Cookies.